

σ_{BF} vs γ^5 disambiguation cleanup — A_sub morning docs

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σ_{BF} vs γ^5 disambiguation cleanup

1. The conflation identified (per Cal #29 first formal application)

When v0.7 edge enumeration applied Cal #29 question-shape audit at design level, the audit surfaced a notation conflation across morning A_sub docs: “ γ^5 ” was used to denote TWO DISTINCT operators:

Operator	Domain	Eigenvalues	Standard relations
σ_{BF} (Pin(2) Z_2 sublattice grading)	All Wallach K-types	+1 (boson, integer m), -1 (fermion, half-integer m)	Commutates with T, C, P (preserves particle type)
γ^5 (Dirac chirality)	Fermion K-types only	+1 (right Weyl), -1 (left Weyl)	Anti-commutes with T, C; commutes with γ^5 -containing P_op (standard Dirac algebra)

These are physically distinct: - σ_{BF} distinguishes bosons from fermions (spin-statistics) - γ^5 distinguishes L-handed from R-handed Weyl spinors within the fermion sublattice

In standard QFT formalism: - σ_{BF} is the SPIN-STATISTICS Z_2 — bosons (integer spin) vs fermions (half-integer spin); T-, C-, P-invariant - γ^5 is the DIRAC CHIRALITY MATRIX — operating on 4-component Dirac spinors; anti-commutes with γ^0 which appears in T-conjugation

The morning A_sub work used “ γ^5 ” for both contexts without explicit distinction.

2. Cleanup status per document

2.1 A_sub v0.4 Section 2.2 — APPLIED

The “Regular form, odd gaps” table identifies the K-type sublattice partition. Disambiguation note added inline: this table uses σ_{BF} (Pin(2) Z_2 sublattice grading) for the boson/fermion distinguisher. v0.7 Section 2 referenced. Edit applied 2026-05-26 PM.

2.2 A_sub commutators steps 3-5 batched doc Section 1.1 — APPLIED

The $\hat{\gamma}^5$ operator definition table identified Pin(2) Z_2 grading as $\varepsilon_K = \pm 1$ for integer/half-integer K-types. Disambiguation note added: this is σ_{BF} , NOT γ^5 -as-Dirac-chirality. The algebraic relations computed in Sections 2-5 (steps 3-5 commutators) — $\{\hat{\gamma}^5, \hat{T}\} = 0$, $\{\hat{\gamma}^5, \hat{C}\} = 0$, $[\hat{\gamma}^5, \hat{P}_{\text{op}}] = 0$ — are for γ^5 -as-Dirac-chirality, per Cal #132 SVC verification. The K-type sublattice identification at the start of Section 1.1 uses σ_{BF} . Edit applied 2026-05-26 PM.

2.3 Half-Integer Axis G v0.2 Section 7.2 — APPLIED

The substrate-natural-spin-statistics observation in Section 7.2 identifies integer K-type sublattice (bosons) and half-integer (fermions) via “ $\hat{\gamma}^5 = \pm 1$ ”. Disambiguation note added: this is σ_{BF} , not γ^5 -as-Dirac-chirality. v0.7 Section 2 referenced. Edit applied 2026-05-26 PM.

2.4 A_sub v0.5 phase-tagging Section 2.3 — minor revision needed

The phase-tagging table refers to “ $\hat{\gamma}^5 \times$ discrete symmetries” — this is γ^5 -as-Dirac-chirality (the operator whose anti-commutation with T, C provides COMPOSITION + COMBINATORIAL phase activation per CPT preservation). The Z_2 sublattice grading appears in Section 3.2 K-type-to-phase mapping (“integer K-types = bosons, half-integer = fermions”) which IS σ_{BF} .

Status: explicit edit deferred to v0.8 phase-tagging update; v0.5 doc remains structurally correct under current interpretation ($\hat{\gamma}^5$ in commutator-tagging context = γ^5 -as-Dirac-chirality; sublattice partition = σ_{BF}). The substantive content of v0.5 is unaffected by the disambiguation.

2.5 A_sub v0.6 multi-phase quiver framework Section 4 — minor revision needed

Section 4 absorbs Grace INV-5180 chirality-inversion finding (“Bosons $\rightarrow$ half-integer Bergman ρ -weights; Fermions $\rightarrow$ integer Bergman ρ -weights”). The “chirality” here refers to the σ_{BF} Z_2 sublattice grading at K-type highest-weight level, transformed under Pin(2) cover bridge to Bergman ρ -weight level.

Status: explicit edit deferred to v0.8 quiver framework update; v0.6 doc remains structurally correct under current interpretation. The “Pin(2) cover bridge as level-translation operator” content stands.

2.6 A_sub v0.7 edge enumeration Section 2 — origin of cleanup; no edit needed

v0.7 Section 2 is the disambiguation source. No edit.

2.7 Step 1-2 + steps 6-8 + step 9 commutator docs — no cleanup needed

These docs do not invoke $\hat{\gamma}^5$ as Z_2 grading operator (they treat $\hat{\gamma}^5$ as Dirac chirality consistently or do not invoke it directly). No edits needed.

3. Cal #132 SVC verifications stand at γ^5 -as-Dirac-chirality interpretation

Cal #132 cold-read verified 8 of 9 morning commutators at SVC tier. The verification covers:

Commutator	Interpretation	SVC interpretation context
$\{\hat{Q}, \hat{P}_{\text{op}}\} = 0$	$\hat{Q} + \hat{P}_{\text{op}} = \gamma^5 \sigma$ (Dirac chirality $\times$ Möbius)	Standard QFT algebra
$[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}]$	Substrate-tick $\times$ Casimir	Model-dependent; Cal noted
$\{\hat{\gamma}^5, \hat{T}\} = 0$	γ^5 Dirac chirality anti-commutes with T-conjugation	Standard Dirac algebra
$\{\hat{\gamma}^5, \hat{C}\} = 0$	γ^5 Dirac chirality anti-commutes with C-conjugation	Standard Dirac algebra
$[\hat{\gamma}^5, \hat{P}_{\text{op}}] = 0$	γ^5 Dirac chirality commutes with $\hat{P}_{\text{op}} = \gamma^5 \sigma$	Standard (P_{op} contains γ^5)
$[\hat{B}, \hat{Q}] = 0$	Bell-CHSH vacuum-localized; charge-blind	Standard projection algebra
$[\hat{L}_i, \hat{\gamma}^5] = 0$	L_i orbital momentum commutes with γ^5 Dirac chirality	Standard (orbital $\times$ spinor independent)
$[\hat{C}_3, \hat{I}_3] = 0$	Gauge factor commutation	Standard direct-product
$[\hat{B}, \hat{T}_{\text{tick}}]$	Bell $\times$ substrate-tick (FRAMEWORK-PLUS)	Multi-week derivation pending

All 8 SVC closures verified for γ^5 -as-Dirac-chirality interpretation. σ_{BF} (Pin(2) Z_2 sublattice grading) is a SEPARATE operator in A_sub whose commutators with T, C, P (commuting per particle-type-preservation) have NOT been computed in

morning work but follow trivially from σ_{BF} 's diagonal eigenvalue structure (the sublattice classification doesn't change under T, C, P).

4. Substrate-CPT theorem disambiguation

Per morning steps 3-5 Section 5.2: $[\gamma^5, \Theta_{\text{CPT}}] = 0$ — substrate-CPT theorem at algebra level.

Disambiguation: this is γ^5 -as-Dirac-chirality commuting with CPT-conjugate operator Θ_{CPT} . Standard Dirac algebra result (T+C signs cancel; P_{op} contains γ^5). The substrate's CPT structure operates on Dirac chirality.

For σ_{BF} (Pin(2) Z_2 sublattice grading): $[\sigma_{\text{BF}}, \Theta_{\text{CPT}}] = 0$ trivially (since σ_{BF} commutes with T, C, P individually). CPT preserves particle type universally — consistent with standard QFT.

Two distinct CPT relations: - γ^5 CPT relation: composite (T+C anti-commute; P commutes); preserved via sign cancellation - σ_{BF} CPT relation: individual T, C, P each commute; trivially preserved

Both stand. They represent CPT preservation at two different algebraic levels.

5. Substrate-CP violation source disambiguation

Per morning steps 3-5 Section 5.3: $\{\gamma^5, \text{CP}\} = 0$ — substrate-algebraic source of observable CP-violation.

Disambiguation: this is γ^5 -as-Dirac-chirality anti-commuting with CP composite. Standard Dirac algebra; T+C signs ANTI-commute; P_{op} commutes; composite anti-commutes overall.

For σ_{BF} : $[\sigma_{\text{BF}}, \text{CP}] = 0$ (commutes; particle-type preserved).

The CP-violation source is at γ^5 -as-Dirac-chirality level, NOT at σ_{BF} level. CP-violation in chirality-resolved measurements (CKM matrix, Kobayashi-Maskawa, etc.) projects from γ^5 algebra; spin-statistics (σ_{BF}) is CP-preserved.

This is structurally consistent with standard QFT: CP-violation in chirality-asymmetric processes (weak interactions) but NOT in spin-statistics (which is preserved by all discrete symmetries).

6. v0.8+ formal incorporation plan

1. Update v0.5 phase-tagging to use $\sigma_{\text{BF}} + \gamma^5$ explicit notation throughout
2. Update v0.6 multi-phase quiver framework Section 4 chirality-inversion to use σ_{BF} notation explicitly (Grace INV-5180 pattern was at σ_{BF} level on K-type sublattice; chirality-inversion-via-Bergman- ρ -translation is the σ_{BF} level-translation operation)
3. Update A_{sub} generator list to include σ_{BF} as separate generator from γ^5 ; total 15 (was 14)

4. Multi-week cleanup: ensure all future A_sub documentation uses σ_{BF} / γ^5 disambiguation explicitly

7. Honest scope

What's RATIFIED / SVC (unaffected by cleanup): - All 8 Cal #132 SVC commutator closures (γ^5 -as-Dirac-chirality interpretation) - Substrate-CPT theorem at γ^5 level - Substrate-CP-violation source at γ^5 level - Step 9 [$\hat{B}$, $\hat{T}_{\text{tick}}$] FRAMEWORK-PLUS

What's FRAMEWORK (with disambiguation applied): - Bose-Fermi K-type sublattice partition (now σ_{BF} ; was conflated with γ^5) - Spin-statistics structurally enforced (σ_{BF} ; standard QFT spin-statistics theorem at substrate level) - Pin(2) chirality-inversion empirical pattern (Grace INV-5180): boson integer $m_1, m_2 \rightarrow$ half-integer Bergman ρ ; fermion half-integer $m_1, m_2 \rightarrow$ integer Bergman ρ — this is σ_{BF} level-translation through Pin(2) cover bridge, not γ^5 inversion

What's NOT affected by cleanup: - Multi-phase quiver math-object candidate (v0.6) — operates at K-type graph level, both σ_{BF} and γ^5 as quiver-node attributes - Phase region classifier (v0.5) — operates at observable level, both operators contribute to discrete-symmetry phase activation - Track DC mechanism candidate (b) — uses $\{\hat{Q}, \hat{P}_{\text{op}}\} = 0$ simultaneous-diagonalizability, doesn't depend on disambiguation

Cal #27 STANDING reflexive trigger count: 0 triggers this doc (operator-identity cleanup; not substantive new claim).

Cal #29 application validation: this cleanup is the direct consequence of Cal #29 question-shape audit at v0.7 design level. Without Cal #29, the conflation would have compounded across v0.5, v0.6, v0.7 + future docs. Cal #29 worked as designed to catch this within 6 minutes of Cal #135 PASSING the candidate.

8. Cleanup status summary

Doc	Status	Action taken
v0.4 Section 2.2	APPLIED	Inline σ_{BF} clarification + reference to v0.7 Section 2
Steps 3-5 batched Section 1.1	APPLIED	Inline σ_{BF} vs γ^5 disambiguation + Cal #132 interpretation note
Half-Integer Axis G v0.2 Section 7.2	APPLIED	σ_{BF} substitution + reference to v0.7 Section 2
v0.5 Section 2.3 + 3.2	DEFERRED to v0.8	Structurally correct under current interpretation

Doc	Status	Action taken
v0.6 Section 4	DEFERRED to v0.8	Structurally correct under current interpretation
v0.7 Section 2	ORIGIN	No edit needed
Steps 1-2, 6-9 docs	NO CLEANUP NEEDED	Don't invoke $\sigma_{\text{BF}}/\gamma^5$ ambiguously

3 docs edited inline + 2 docs flagged for v0.8 future edit + 5 docs unaffected. Total cleanup time ~30 minutes.

9. Coordination

Cal: validates cleanup; Cal #132 SVC verifications stand at γ^5 -as-Dirac-chirality interpretation. v0.7 Section 2 disambiguation flag was Cal #29 first formal application; cleanup is the operational consequence.

Keeper: integration into Vol 15 Ch 9 case study draft — note that disambiguation is operator-identity cleanup, not cascade-failure (per Keeper Priority 1 directive confirmation).

Grace: INV-5180 chirality-inversion pattern reinterpreted at σ_{BF} level — Pin(2) cover bridge translates σ_{BF} eigenvalue between K-type highest-weight level (where σ_{BF} aligns with integer vs half-integer) and Bergman ρ -weight level (where σ_{BF} anti-aligns due to ρ -vector half-integer offset). This is structurally clean; substantive content of INV-5180 unaffected.

Elie: K-type sublattice partition in Toy 3535/3537 (12 bosons + 16 fermions + 0 mixed-forbidden) is at σ_{BF} level. Toy 3538 Dirac-ground-state forward derivation would operate at γ^5 -as-Dirac-chirality level (within fermion sublattice). Disambiguation supports clean future toy design.

— Lyra, σ_{BF} vs γ^5 disambiguation cleanup v0.1 filed Tuesday 2026-05-26 ~10:25 EDT per Keeper Priority 1 directive + Casey continue-Lyra-pulls authorization. v0.1 CLEANUP grade (not new substantive content; operator-identity hygiene). 3 docs edited inline; 2 deferred to v0.8 (structurally correct under current interpretation); 5 unaffected. Cal #132 SVC verifications stand at γ^5 -as-Dirac-chirality interpretation. Cal #29 first formal application validated: caught the conflation within 6 minutes of candidate filing.